Sample data : a sample from a large population
Population data : all data
Deming experiment
Process should not be adjusted in response to random variation.
Variance
average physical distance from mean
s^2 = VAR.S(data)
Standard deviation
s = STEV.S(data) = \sqrt(s^2)
Mean Median Mode
AVERAGE MEDIAN MODE(range)
Percentile
K% (percentile) of all data values are less than it. 80% of the data values lie below 80th percentile.
we use exclude as the main one, percentile.exc
Include is population, exclude is sample
Quartile
25 percentile amount for each quartile
Range and inter quartile range
Inter quartile range = Q3 - Q1
Range = max - min
Stem and leaf diagram
Leaf must always be a single digit
Ordered stem and leaf just have ordered leaf
Histogram
Bar chart
Pareto chart
similar to bar chart I guess
Box plot

Chapter 2 need to know for cheat sheet
Box plot : Upper limit = 1.5IQR + Q3
Lower limit = 1.5IQR - Q1
correl(data A, data B) finding correlation between two data, a plot can be created as well
upper quartile != upper limit
Probability
P(A|B) = P(A sec B) / P(B)
Order matters = factorial
Order does not matter = Out of n total items, I am choosing r items.
n fact / r fact (n-r) fac
Norm.dist, norm.s.dist
Standard normal distribution is a normal distribution with 0 mean and 1 variance
continuous btw
From normal variable to standard norm variable
Z = (x-mean)/standard deviation
P(Z, z) means standard normal distribution z
For standard normal distribution, P(Z<4.5) and P(Z<=4.5) are the same
Inv converts from standard normal distbution chance to a Z (random var)
Uniform distribution
mean = sum of all x f(x)
standard deviation = sum of x^2 F(x) * mean ^ 2
power(a, b) = a^ b
short cut : in case that all f(x) of all x values are equal,
mean = 1/2 * (min+max)
variance = (max-min+1)^2-1)/12
sd = sqrt(variance)
Binom.dist
Typically a discrete function, but have cumulative as well
binom.dist(random_var, no. of trials, prob of success, cumulative?)
=binom.dist(number of success we want, number of total trials that will happen, prob of success per one occurence, cumulative?)
=POISSON.DIST(number of occurrences we want, average rate of occurrences that will happen, cumulative?)
Poisson lacks hard limit
Binom and poisson
Binom measures the number of success of hard limit "n" trials
Whilst poisson counts a number of success over a continuous field or window (time, distance, area) no fixed upper limit of how many times it can happen
binom is discrete, poisson is continuous
Normal distribution
All you need is there, bell curve
Binom to normal.s
convert from x to z using formulas x-mean/(sqrt(n(1-p)(p))
'Z=(X+-0.5-mean)/std (x adds 0.5 due to convert from continuous to discrete
P(X = 5) ≈ P(4.5<=X<=5.5)
std = "=SQRT(np(1-p))
np=mean
Confidence interval
alpha = 1- confidence interval
z=alpha/2

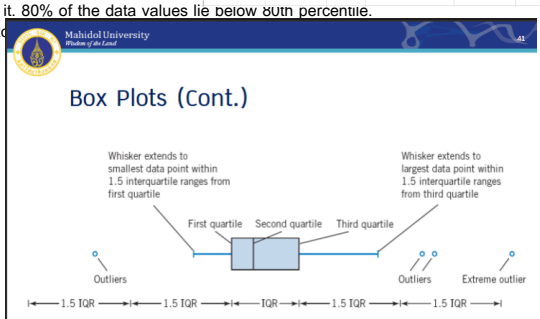
Sample	Energy
1	64.1
2	64.7
3	64.5
4	64.6
5	64.5
6	64.3
7	64.6
8	64.8
9	64.2
10	64.3

If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2, a $100(1-\alpha)\%$ CI on μ is given by

$$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \le \mu \le \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$$

where $z_{\alpha/2}$ is the upper $100\alpha/2$ percentage point of the standard normal distribution.

mean (x-bar)	64.46	95% CI	=100(1- alpha)% = 95
z	1.959964	=NORM.S.INV(1-0.05/2)	alpha = 0.05
s.d. (sigma)	1	0.316228	alpha/2 = 0.025
Lower Bound	63.8402	=E9-E10*F11	
Upper Bound	65.0798	=E9+E10*F11	



Description of a box plot

A $100(1-\alpha)\%$ **upper-confidence bound** for μ is

$$\mu \le u = \bar{x} + z_{\alpha}\sigma/\sqrt{n}$$

and a $100(1-\alpha)\%$ **lower-confidence bound** for μ is

$$\bar{x} - z_{\alpha}\sigma/\sqrt{n} = l \le \mu$$

z	z(alpha)	1.644854
Upper Confidence Bound	64.98015	

z	z(alpha)
Lower Confidence Bound	63.93985

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

E	0.5	
n ≈	15.36584	=POWER((E10*1)/E25,2)
∴ n =	16	

	Name	Height
1	Techo	173
2	Lavish	166
3	Kim	180
4	Matt	185
5	Pin	177
6	Kim	168
7	Earth	154
8	Phufah	175
9	Thann	177
10	Fluke	175
11	Fuse	171
12	Win	171
13	Noon	168
14	Jean	164
15	Amy	163
16	KJ	178
17	Phu	170
18	Aki	170
19	Phin	168
20	Bun	180
21	Amro	170

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then **a $100(1-\alpha)\%$ confidence interval on σ^2 is**

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom, respectively. A **confidence interval for σ** has lower and upper limits that are the square roots of the corresponding limits in this equation.

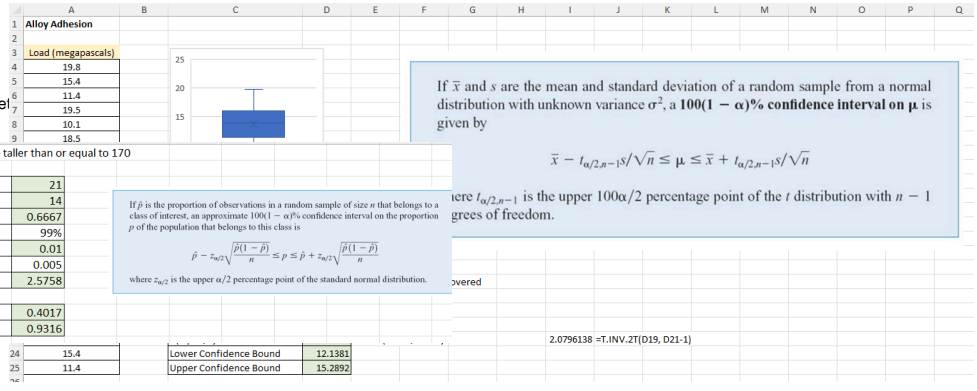
Sample Size	21	B2 -> B22
Sample Variance (s^2)	48.2571	
Confident Interval (CI)	99%	
Alpha (α)	0.01	
Alpha/2 (α/2)	0.005	
d.f.	20	
Chi-square (χ^2) (alpha/2)	39.9968	Chi-square (c^2)(1-alpha/2) 7.43384

Lower Confidence Bound of Variance	24.1305
Upper Confidence Bound of Variance	129.8309
Lower Confidence Bound of Standard Deviation	4.9123
Upper Confidence Bound of Standard Deviation	11.3943

SUM =CHISQ.INV.RT(1-B6, B7)

	A	B	C	D
1	**Detergent Filling**			
2				
3	Sample Size	20		
4	Sample Variance (s^2)	0.0153		
5	Confident Interval (CI)	95%		
6	Alpha (α)	0.05		
7	d.f.	19		
8	Chi-square (χ^2)	RT(1-B6, B7)		

The 10
respect



If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a **$100(1-\alpha)\%$ confidence interval on μ** is given by

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \le \mu \le \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n}$$

ere $t_{\alpha/2,n-1}$ is the upper $100\alpha/2$ percentage point of the t distribution with $n-1$ grees of freedom.

	Name	Height
1	Techo	173
2	Lavish	166
3	Kim	180
4	Matt	185
5	Pin	177
6	Kim	168
7	Earth	154
8	Phufah	175
9	Thann	177
10	Fluke	175
11	Fuse	171

Nmber of students that are taller than or equal to 170

Sample Size (n)	21
# Success (x)	14
Sample Proportion	0.6667
Confident Interval (CI)	99%
Alpha (α)	0.01
Alpha/2 (α/2)	0.005
z(α/2)	2.5758
Lower Confidence Bound	0.4017
Upper Confidence Bound	0.9316

If $\hat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1-\alpha)\%$ confidence interval on the proportion p of the population that belongs to this class is

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

overed

2.0796138 =T.INV.2T(D19, D21-1)

Lower Confidence Bound	12.1381
Upper Confidence Bound	15.2892

Crankshaft Bearings

Sample Size (n)	85
# Success (x)	10
Sample Proportion	0.1176
Confident Interval (CI)	95%
Alpha (α)	0.05
Alpha/2 (α/2)	0.025
z(α/2)	1.9600
Lower Confidence Bound	0.0492
Upper Confidence Bound	0.1861
z(alpha)	1.6448536

If $\hat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1-\alpha)\%$ confidence interval on the proportion p of the population that belongs to this class is

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

The approximate $100(1-\alpha)\%$ lower and upper confidence bounds are

$$\hat{p} - z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \quad \text{and} \quad p \le \hat{p} + z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

respectively.

z(α)	1.6449
Lower Confidence Bound	0.0602

z(α)	1.6449
Upper Confidence Bound	0.1751